

1 Mask

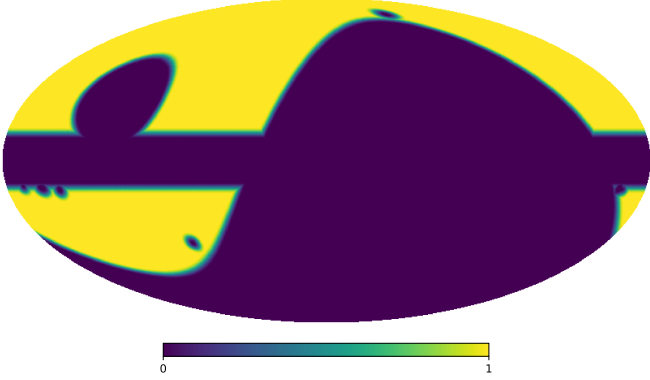


Figure 1: Analysis mask combining the QUIJOTE low-declination survey mask with a 10 mK intensity threshold applied to the QUIJOTE 11 GHz map. The mask excludes regions with high foreground contamination and point sources to ensure reliable polarimetric analysis of diffuse Galactic emission.

2 Data

Our analysis employs two complementary datasets: simulated sky maps generated using the `Python Sky Model (PySM)` library, which incorporates all major Galactic foreground components; and observational data from the QUIJOTE, WMAP, and Planck experiments. All simulated maps are convolved with the appropriate beam window functions for each respective experiment to ensure realistic instrumental effects.

Before fitting the observed power spectra, we subtract the CMB contribution using the best-fit theoretical Λ CDM spectrum from Planck 2018 (Planck Collaboration, 2020). This spectrum is obtained from the full Plik TTTEEE likelihood combined with low-multipole (lowE) and gravitational lensing constraints. For polarization modes, we subtract both C_ℓ^{EE} and C_ℓ^{BB} , where the latter includes only the contribution from B-modes generated by gravitational lensing, since primordial B-modes of inflationary origin have not yet been detected. This CMB subtraction is essential to isolate the Galactic foreground emission (synchrotron and thermal dust) that constitutes the target of this analysis.

3 Error estimation

Power spectrum uncertainties are estimated through Monte Carlo simulations for both datasets. We compute the standard deviation from 100 realizations each of sky+noise and pure noise maps. The noise characteristics vary by experiment: QUIJOTE and Planck employ the official noise simulations provided by their respective collaborations, incorporating both white noise and correlated $1/f$ noise components. For WMAP, we model the noise as purely white, following standard practice for this dataset.

4 MCMC fitting

The total angular power spectrum between frequencies ν_i and ν_j is modeled as

$$C_\ell(\nu_i, \nu_j) = C_\ell^{\text{sync}}(\nu_i, \nu_j) + C_\ell^{\text{dust}}(\nu_i, \nu_j) + C_\ell^{\text{rsd}}(\nu_i, \nu_j), \quad (1)$$

where the synchrotron, dust, and synchrotron-dust cross-correlation terms are

$$C_\ell^{\text{sync}}(\nu_i, \nu_j) = A_s \left(\frac{\ell}{\ell_{\text{ref}}} \right)^{\alpha_s} \left(\frac{\nu_i}{\nu_{\text{ref}}} \right)^{\beta_s} \left(\frac{\nu_j}{\nu_{\text{ref}}} \right)^{\beta_s}, \quad (2)$$

$$C_\ell^{\text{dust}}(\nu_i, \nu_j) = A_d \left(\frac{\ell}{\ell_{\text{ref}}} \right)^{\alpha_d} S_d(\nu_i) S_d(\nu_j), \quad (3)$$

$$C_\ell^{\text{rsd}}(\nu_i, \nu_j) = \rho \sqrt{A_s A_d} \left[S_s(\nu_i) S_d(\nu_j) + S_s(\nu_j) S_d(\nu_i) \right] \left(\frac{\ell}{\ell_{\text{ref}}} \right)^{(\alpha_s + \alpha_d)/2}, \quad (4)$$

with $S_s(\nu) = (\nu/\nu_{\text{ref}})^{\beta_s}$ and the modified black-body scaling in Rayleigh-Jeans units

$$S_d(\nu) = \frac{B_\nu(\nu, T_d)}{B_\nu(\nu_0, T_d)} \left(\frac{\nu}{\nu_0} \right)^{\beta_d}, \quad (5)$$

where B_ν is the Planck function. Here, A_s and A_d are the amplitudes of the synchrotron and dust components at the reference multipole $\ell_{\text{ref}} = 80$ and frequencies $\nu_{\text{ref}} = 11.1$ GHz and $\nu_0 = 353.0$ GHz, respectively; α_s and α_d are the multipole spectral indices; β_s and β_d are the frequency spectral indices; T_d is the dust temperature (fixed at $T_d = 19.6$ K); and ρ is the synchrotron-dust correlation coefficient.

4.1 Data Selection and Frequency Coverage

We perform a Markov Chain Monte Carlo (MCMC) fitting procedure using data from QUI-JOTE 11 GHz; WMAP 23, 33, 41, 61, and 94 GHz; and Planck 30, 44, 70, 100, 143, 217, and 353 GHz, covering the multipole range $30 < \ell < 200$. In this study, we use both auto- and cross-spectra between frequency bands. This yields 91 independent spectra from the 13 frequency bands.

4.2 Bayesian Inference

We adopt a Bayesian framework to constrain the model parameters $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$. The posterior distribution is given by

$$p(\boldsymbol{\theta} \mid \mathbf{D}) \propto \mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}), \quad (6)$$

where $\mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})$ is the likelihood and $p(\boldsymbol{\theta})$ is the prior. The likelihood is constructed from the observed power spectra $\hat{C}_\ell(\nu_i, \nu_j)$ and their uncertainties:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i,j,\ell} \frac{[\hat{C}_\ell(\nu_i, \nu_j) - C_\ell(\nu_i, \nu_j; \boldsymbol{\theta})]^2}{\sigma_\ell^2(\nu_i, \nu_j)}. \quad (7)$$

For the power-law model fitting, we adopt uniform (flat) priors for all model parameters within physically motivated bounds:

$$A_s \in [0, \infty), \quad (8)$$

$$\alpha_s \in [-6, 0], \quad (9)$$

$$\beta_s \in [-6, 0], \quad (10)$$

$$A_d \in [0, \infty), \quad (11)$$

$$\alpha_d \in [-6, 0], \quad (12)$$

$$\beta_d \in [0, 6], \quad (13)$$

$$\rho \in [-1, 1], \quad (14)$$

For the bin-to-bin analysis, to regularize the fit and incorporate external information from the literature, we impose a Gaussian prior on the synchrotron frequency spectral index:

$$\beta_s \sim \mathcal{N}(-3.1, 0.30^2), \quad (15)$$

while maintaining uniform priors for the remaining parameters within the bounds specified above.

4.3 Fitting methods

We apply three complementary approaches to the angular power spectra computed from the maps:

Power-law fitting, $\Delta\ell = 10$ We compute the spectra with a bin width $\Delta\ell = 10$ and fit a power-law model for the multipole dependence of the primary foreground components (synchrotron and dust). This fit estimates the amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), the multipole spectral indices (α_s, α_d), and the correlation coefficient (ρ) between components.

Power-law fitting, $\Delta\ell = 20$ We repeat the power-law fit using a wider binning, $\Delta\ell = 20$, to test the stability of the fitted parameters with respect to the choice of bin size. Wider bins reduce per-bin noise at the cost of multipole resolution.

Bin-to-bin analysis, $\Delta\ell = 20$ In this approach each multipole bin is fitted independently with $\Delta\ell = 20$, without assuming a global power-law dependence on ℓ . For each bin we only fit the amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), and the correlation (ρ); the multipole spectral indices (α_s, α_d) are not constrained in this analysis. This method allows us to identify scale-dependent behaviour and local deviations from the power-law assumption.

5 Results

In the following subsections we first present the results obtained on simulated data and then the results for observational data, applying the three procedures described previously (Section 4.3).

5.1 Simulated data

We compute the angular power spectra from the PySM simulated maps. Below we present the simulated-data results organised by fitting method: power-law fits with $\Delta\ell = 10$ and $\Delta\ell = 20$, and the bin-to-bin analysis with $\Delta\ell = 20$.

5.1.1 Power-law fit ($\Delta\ell = 10$)

The MCMC fitting of the $\Delta\ell = 10$ spectra yields posterior distributions shown in the corner plots of Figure 2 (EE mode) and Figure 3 (BB mode). These plots illustrate the parameter constraints

and correlations for the synchrotron and dust components.

5.1.2 Power-law fit ($\Delta\ell = 20$)

To test stability with respect to binning, the power-law fit is repeated with $\Delta\ell = 20$. The resulting posterior distributions are displayed in Figure 4 (EE) and Figure 5 (BB).

5.1.3 Bin-to-bin analysis ($\Delta\ell = 20$)

The bin-to-bin analysis fits each multipole bin independently (see Section 4.3). Table 1 summarises the fitted parameters per bin, and Figure 6 shows their evolution with multipole ℓ .

5.2 Observational data

We show results for the same set of fitting approaches used on the simulations to allow direct comparison.

5.2.1 Power-law fit ($\Delta\ell = 10$)

The MCMC fitting of the observational spectra with $\Delta\ell = 10$ yields posterior distributions shown in Figure 7 (EE) and Figure 8 (BB).

5.2.2 Power-law fit ($\Delta\ell = 20$)

The power-law fit with wider bins ($\Delta\ell = 20$) is shown in Figure 9 (EE) and Figure 10 (BB).

5.2.3 Bin-to-bin analysis ($\Delta\ell = 20$)

The bin-to-bin analysis for observational data fits each multipole bin independently; results are listed in Table 2 and illustrated in Figure 11.

References

Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *A&A*, 641:A6.

Table 1: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{EE}} [\mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{EE}}$	$A_{\text{dust}}^{\text{EE}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{EE}}$	ρ^{EE}
4–11	7.0	2076.45 ± 199.24	-2.96 ± 0.06	3310.73 ± 139.80	1.55 ± 0.06	-0.221 ± 0.039
12–19	15.0	63.30 ± 6.08	-2.92 ± 0.06	235.56 ± 3.98	1.59 ± 0.02	0.080 ± 0.025
20–39	29.0	20.88 ± 0.69	-2.93 ± 0.02	46.12 ± 0.34	1.53 ± 0.01	-0.058 ± 0.009
40–59	49.0	4.84 ± 0.22	-2.93 ± 0.03	7.64 ± 0.09	1.52 ± 0.01	0.014 ± 0.011
60–79	69.0	2.15 ± 0.12	-2.94 ± 0.04	4.35 ± 0.05	1.54 ± 0.01	-0.036 ± 0.012
80–99	89.0	1.08 ± 0.09	-2.91 ± 0.06	2.00 ± 0.03	1.55 ± 0.02	-0.053 ± 0.016
100–119	109.0	0.61 ± 0.16	-2.99 ± 0.60	1.33 ± 0.05	1.62 ± 0.36	-0.038 ± 0.135
120–139	129.0	0.40 ± 0.08	-2.93 ± 0.14	0.86 ± 0.02	1.54 ± 0.03	-0.030 ± 0.098
140–159	149.0	0.24 ± 0.06	-2.86 ± 0.19	0.56 ± 0.01	1.57 ± 0.04	0.002 ± 0.087
160–179	169.0	0.17 ± 0.07	-3.24 ± 0.24	0.43 ± 0.01	1.51 ± 0.03	0.121 ± 0.077
180–199	189.0	0.20 ± 0.08	-3.15 ± 0.24	0.32 ± 0.01	1.50 ± 0.04	-0.005 ± 0.057
ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{BB}} [\mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{BB}}$	$A_{\text{dust}}^{\text{BB}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{BB}}$	ρ^{BB}
4–11	7.0	1620.94 ± 178.43	-3.12 ± 0.07	1181.63 ± 37.28	1.65 ± 0.05	0.565 ± 0.039
12–19	15.0	25.25 ± 5.04	-2.93 ± 0.30	118.31 ± 2.70	1.49 ± 0.09	-0.060 ± 0.045
20–39	29.0	5.42 ± 0.39	-3.05 ± 0.05	18.38 ± 0.24	1.59 ± 0.02	0.288 ± 0.019
40–59	49.0	1.04 ± 0.13	-2.96 ± 0.09	4.32 ± 0.06	1.53 ± 0.02	0.047 ± 0.023
60–79	69.0	0.47 ± 0.08	-2.99 ± 0.12	2.13 ± 0.03	1.57 ± 0.02	0.207 ± 0.028
80–99	89.0	0.21 ± 0.07	-3.13 ± 0.38	1.00 ± 0.05	1.51 ± 0.18	0.044 ± 0.103
100–119	109.0	0.12 ± 0.05	-3.14 ± 0.39	0.66 ± 0.02	1.59 ± 0.30	0.001 ± 0.058
120–139	129.0	0.15 ± 0.06	-3.26 ± 0.40	0.43 ± 0.01	1.60 ± 0.12	-0.088 ± 0.063
140–159	149.0	0.02 ± 0.02	-3.10 ± 0.27	0.33 ± 0.01	1.58 ± 0.04	-0.375 ± 0.226
160–179	169.0	0.05 ± 0.04	-3.10 ± 0.28	0.26 ± 0.01	1.59 ± 0.04	-0.107 ± 0.179
180–199	189.0	0.01 ± 0.03	-3.22 ± 0.29	0.19 ± 0.01	1.55 ± 0.05	-0.026 ± 0.287

EE Mode

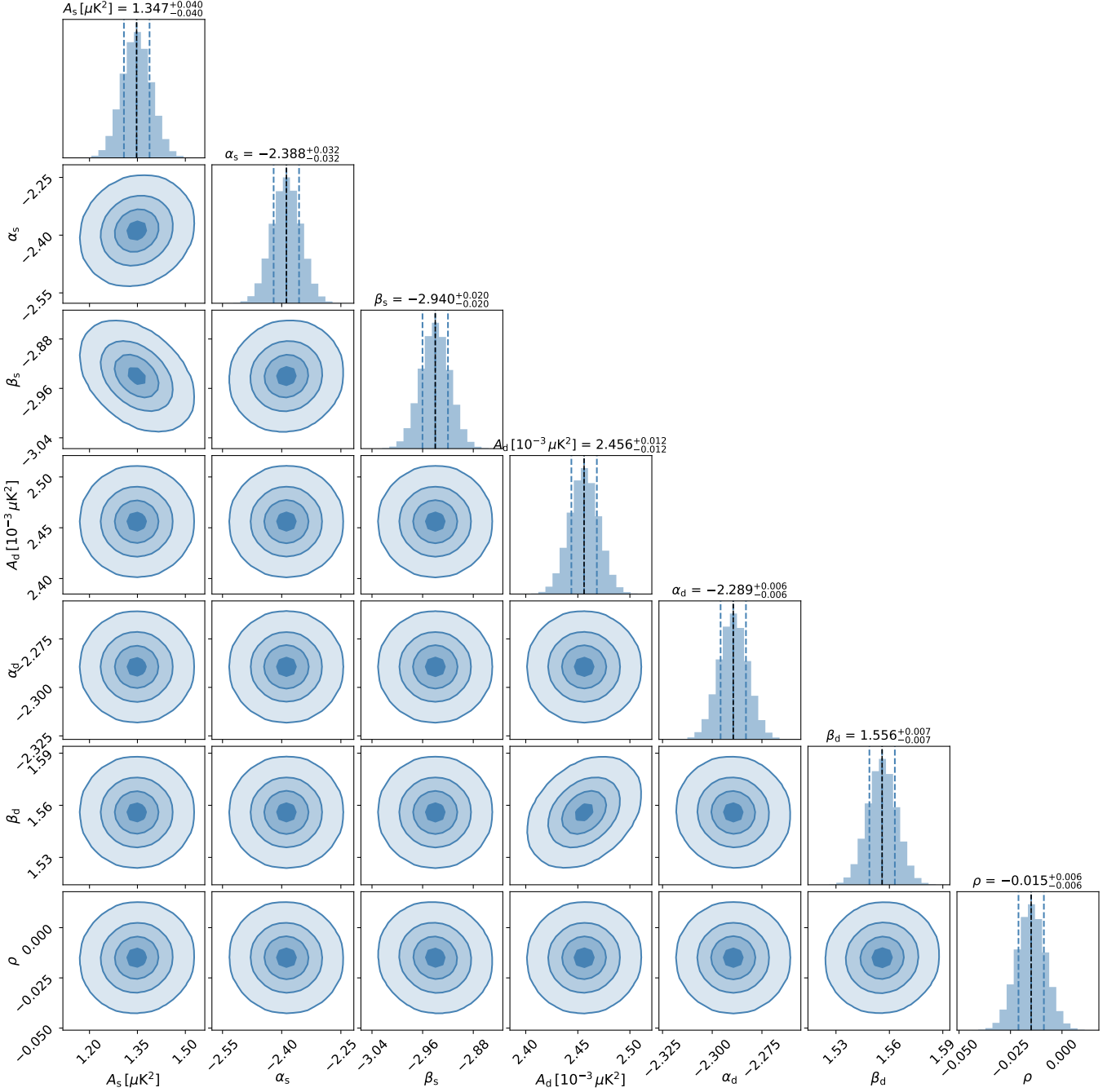


Figure 2: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of simulated EE mode data using $\Delta\ell = 10$.

BB Mode

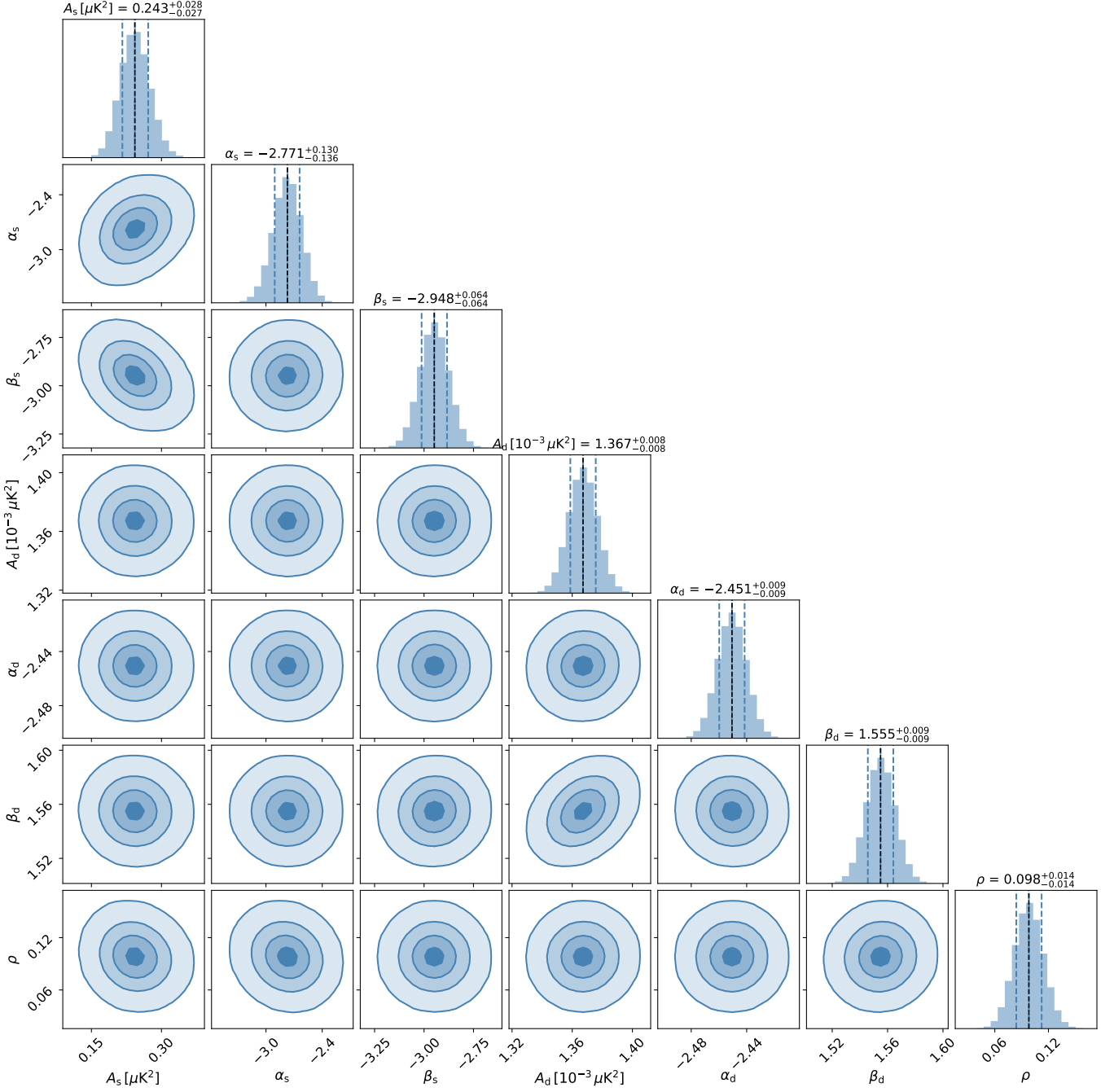


Figure 3: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of simulated BB mode data using $\Delta\ell = 10$.

EE Mode

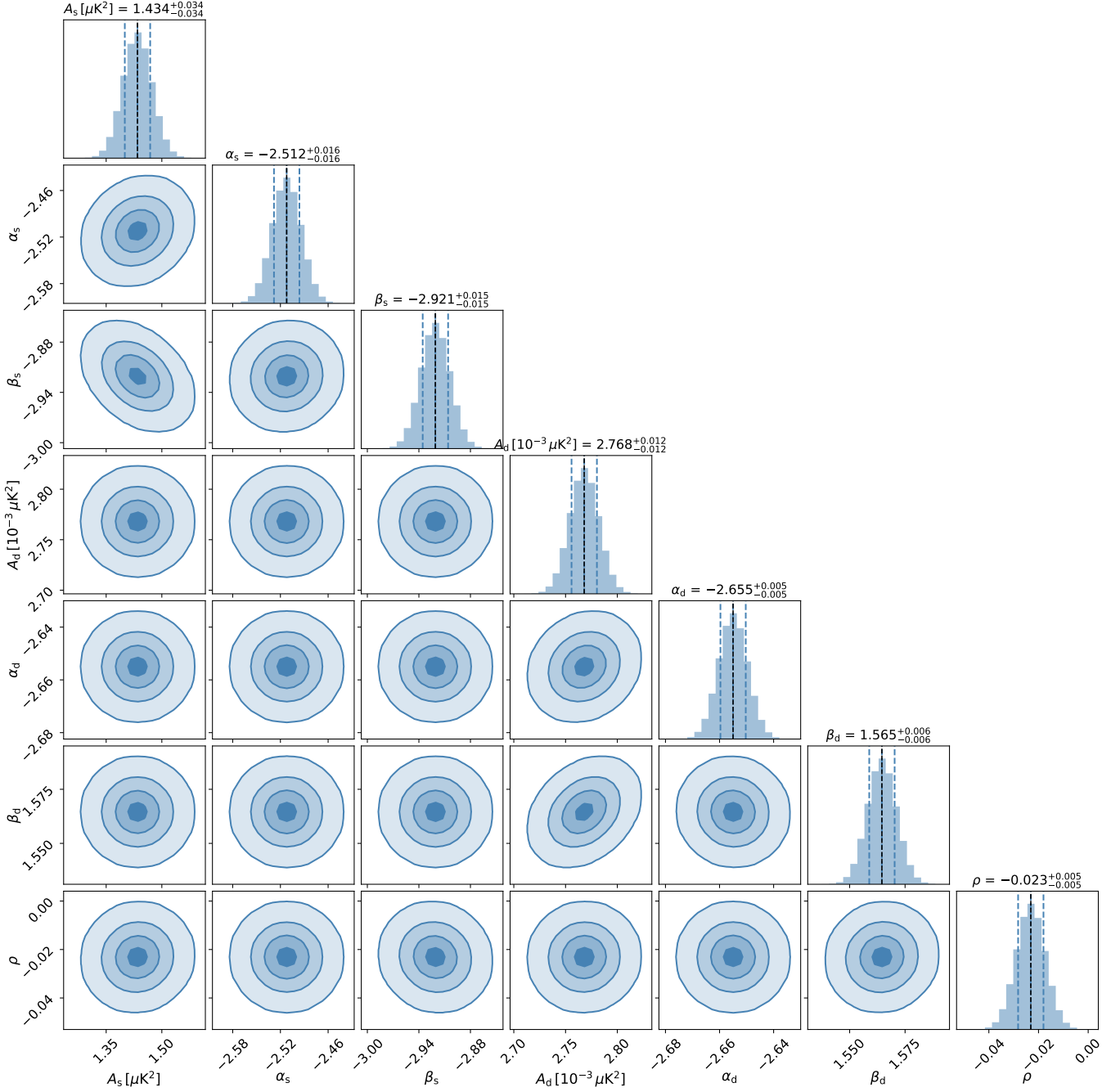


Figure 4: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of simulated EE mode data using $\Delta\ell = 20$.

BB Mode

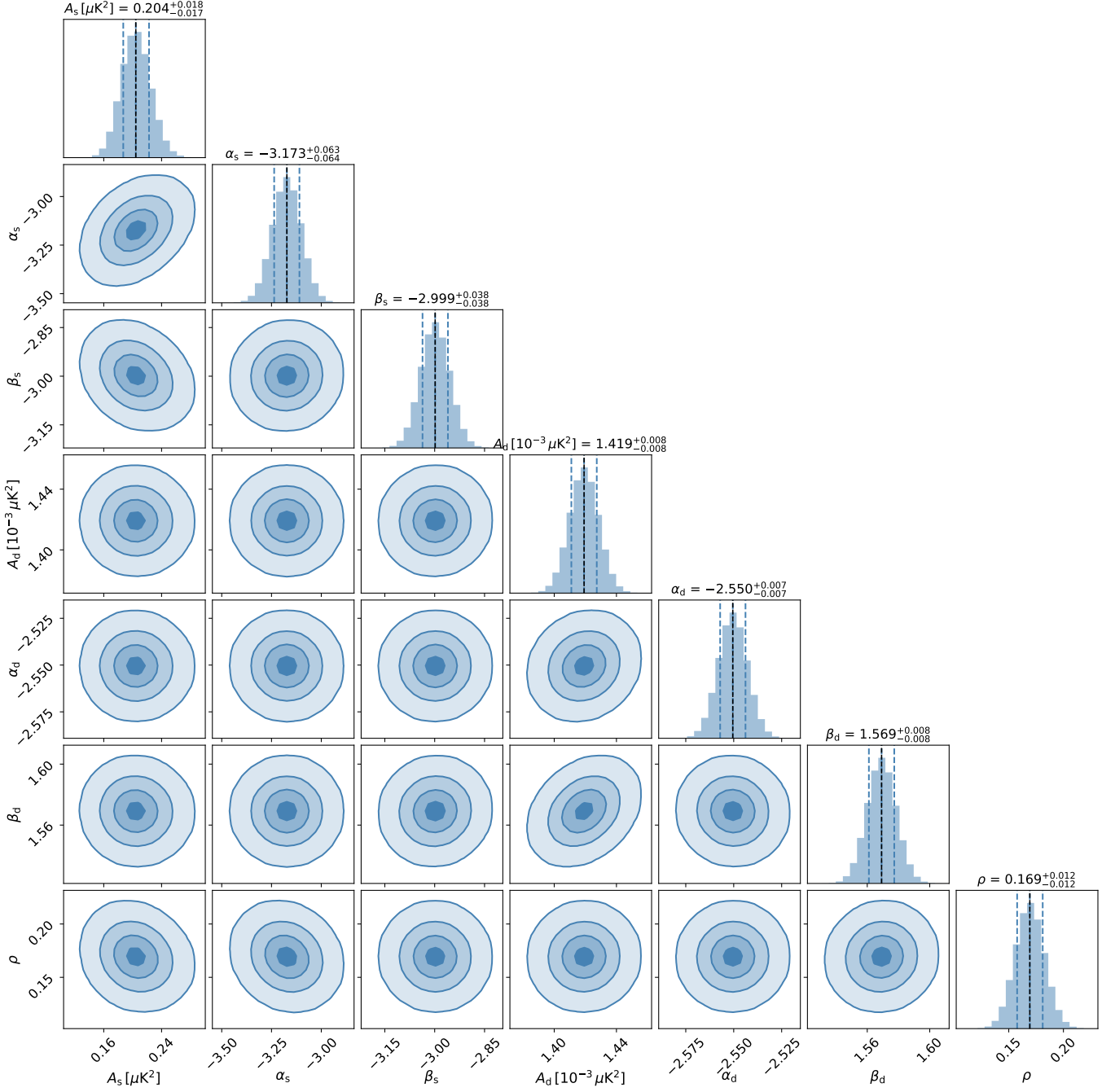


Figure 5: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of simulated BB mode data using $\Delta\ell = 20$.

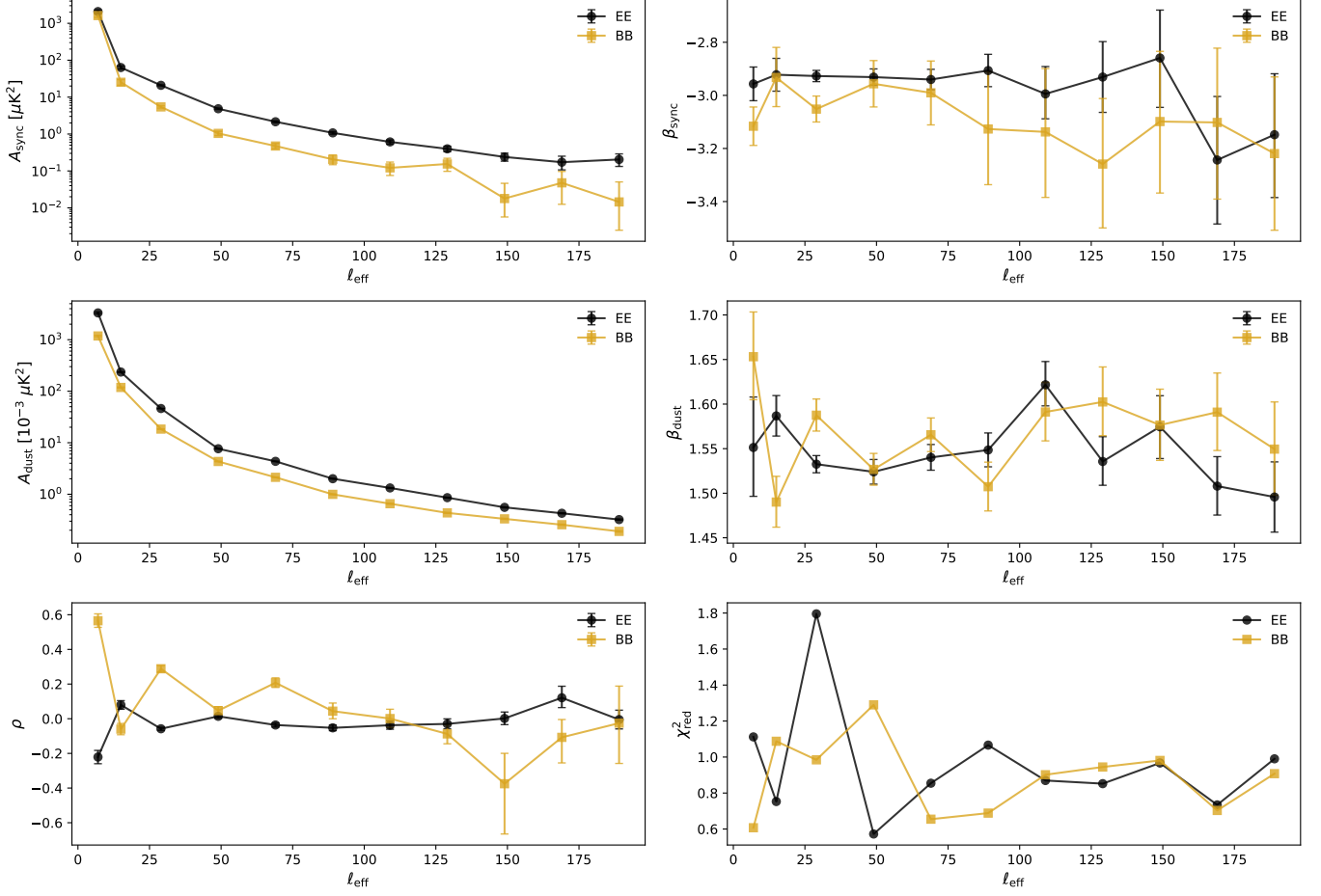


Figure 6: Bin-to-bin analysis results (simulations) showing the evolution of fitted synchrotron and dust parameters as a function of multipole ℓ for both EE (top) and BB (bottom) modes. The reduced χ^2 values for each bin are shown in the bottom panels.

EE Mode

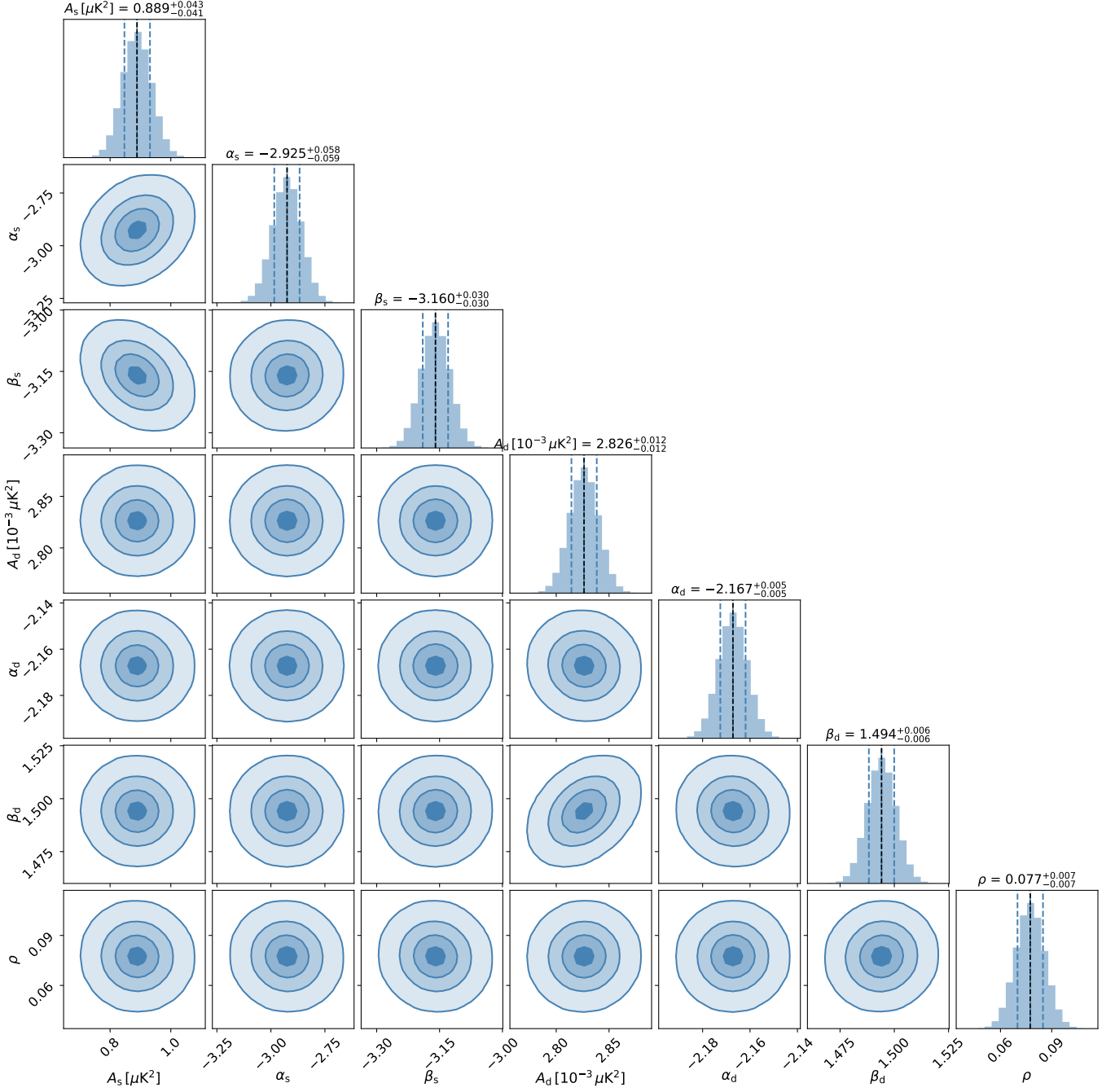


Figure 7: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational EE mode data with bin width $\Delta\ell = 10$.

BB Mode

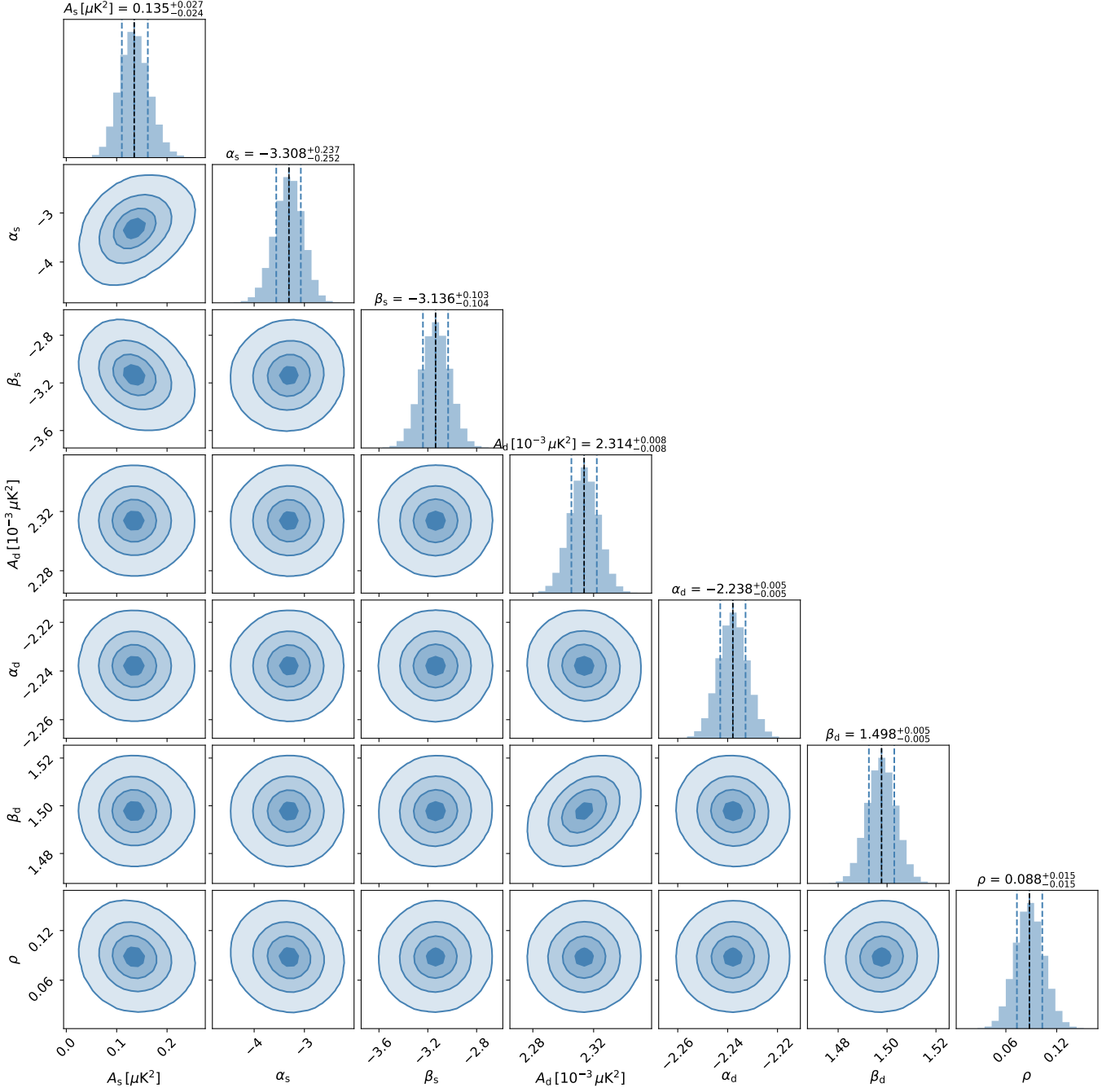


Figure 8: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational BB mode data with bin width $\Delta\ell = 10$.

EE Mode

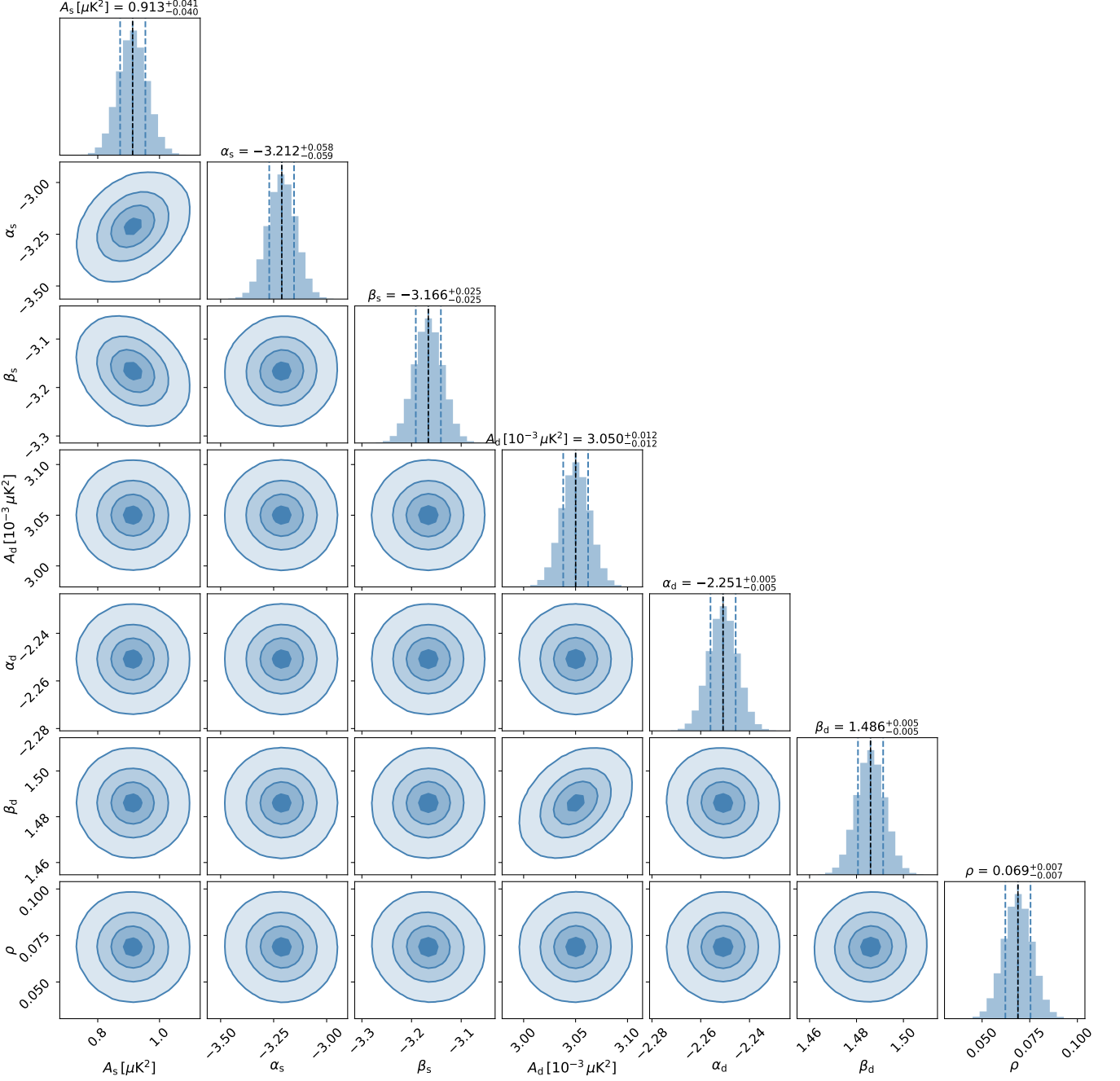


Figure 9: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational EE mode data with bin width $\Delta\ell = 20$.

BB Mode

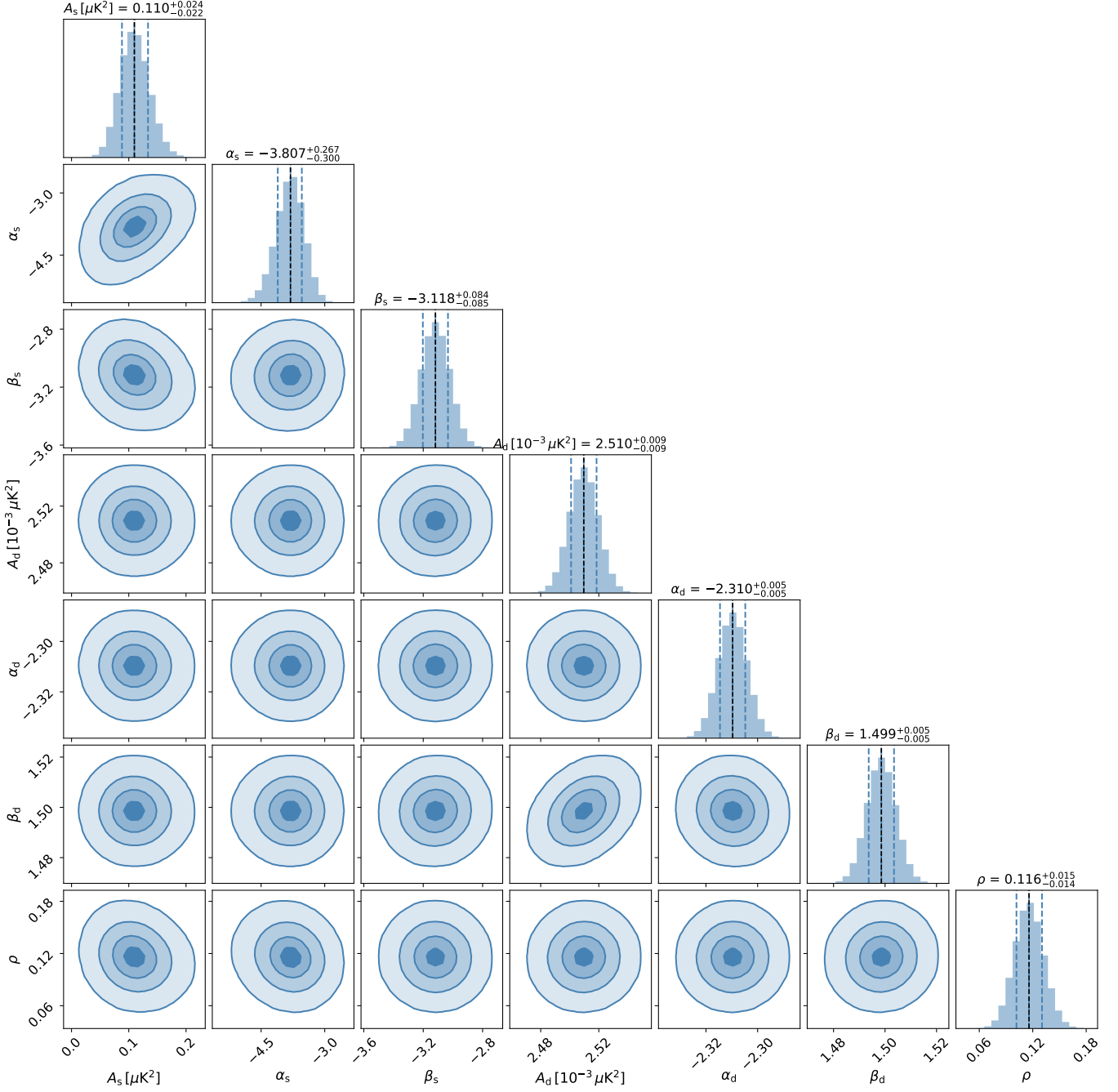


Figure 10: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational BB mode data with bin width $\Delta\ell = 20$.

Table 2: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{EE}} [\mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{EE}}$	$A_{\text{dust}}^{\text{EE}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{EE}}$	ρ^{EE}
4–11	7.0	814.04 ± 189.74	-3.01 ± 0.15	3546.06 ± 136.25	1.49 ± 0.05	0.164 ± 0.061
12–19	15.0	124.31 ± 6.46	-3.04 ± 0.04	338.79 ± 3.83	1.47 ± 0.01	0.169 ± 0.015
20–39	29.0	27.32 ± 0.78	-3.09 ± 0.02	45.06 ± 0.33	1.45 ± 0.01	0.070 ± 0.008
40–59	49.0	4.00 ± 0.25	-3.15 ± 0.05	8.32 ± 0.09	1.50 ± 0.01	0.130 ± 0.013
60–79	69.0	1.60 ± 0.30	-3.26 ± 0.43	4.69 ± 0.07	1.51 ± 0.01	0.035 ± 0.106
80–99	89.0	0.69 ± 0.16	-3.22 ± 0.56	2.63 ± 0.24	1.48 ± 0.24	0.029 ± 0.098
100–119	109.0	0.26 ± 0.08	-3.15 ± 0.19	1.60 ± 0.02	1.55 ± 0.02	0.060 ± 0.125
120–139	129.0	0.13 ± 0.05	-2.81 ± 0.22	1.02 ± 0.01	1.38 ± 0.02	0.201 ± 0.054
140–159	149.0	0.14 ± 0.06	-3.22 ± 0.40	0.70 ± 0.04	1.45 ± 0.08	0.235 ± 0.092
160–179	169.0	0.09 ± 0.07	-3.36 ± 0.42	0.51 ± 0.01	1.51 ± 0.11	0.133 ± 0.151
180–199	189.0	0.05 ± 0.05	-3.21 ± 0.27	0.42 ± 0.01	1.60 ± 0.03	0.251 ± 0.193
ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{BB}} [\mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{BB}}$	$A_{\text{dust}}^{\text{BB}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{BB}}$	ρ^{BB}
4–11	7.0	540.03 ± 119.00	-2.89 ± 0.12	1756.33 ± 37.20	1.50 ± 0.03	0.474 ± 0.053
12–19	15.0	29.58 ± 4.51	-2.86 ± 0.10	171.01 ± 2.59	1.50 ± 0.02	-0.021 ± 0.024
20–39	29.0	3.90 ± 0.37	-3.04 ± 0.07	26.07 ± 0.24	1.49 ± 0.01	0.144 ± 0.017
40–59	49.0	1.13 ± 0.17	-3.26 ± 0.12	6.60 ± 0.06	1.47 ± 0.01	0.058 ± 0.020
60–79	69.0	0.14 ± 0.07	-3.27 ± 0.41	3.40 ± 0.15	1.52 ± 0.20	0.095 ± 0.076
80–99	89.0	0.04 ± 0.04	-3.17 ± 0.24	2.04 ± 0.02	1.52 ± 0.01	0.188 ± 0.177
100–119	109.0	0.02 ± 0.02	-2.85 ± 0.27	1.36 ± 0.01	1.53 ± 0.01	0.182 ± 0.182
120–139	129.0	0.04 ± 0.04	-3.06 ± 0.40	0.84 ± 0.06	1.51 ± 0.12	0.169 ± 0.162
140–159	149.0	0.01 ± 0.02	-3.17 ± 0.43	0.54 ± 0.04	1.48 ± 0.07	-0.103 ± 0.269
160–179	169.0	0.12 ± 0.06	-3.17 ± 0.30	0.42 ± 0.01	1.47 ± 0.02	-0.019 ± 0.071
180–199	189.0	0.08 ± 0.07	-3.10 ± 0.39	0.36 ± 0.01	1.46 ± 0.03	-0.010 ± 0.145

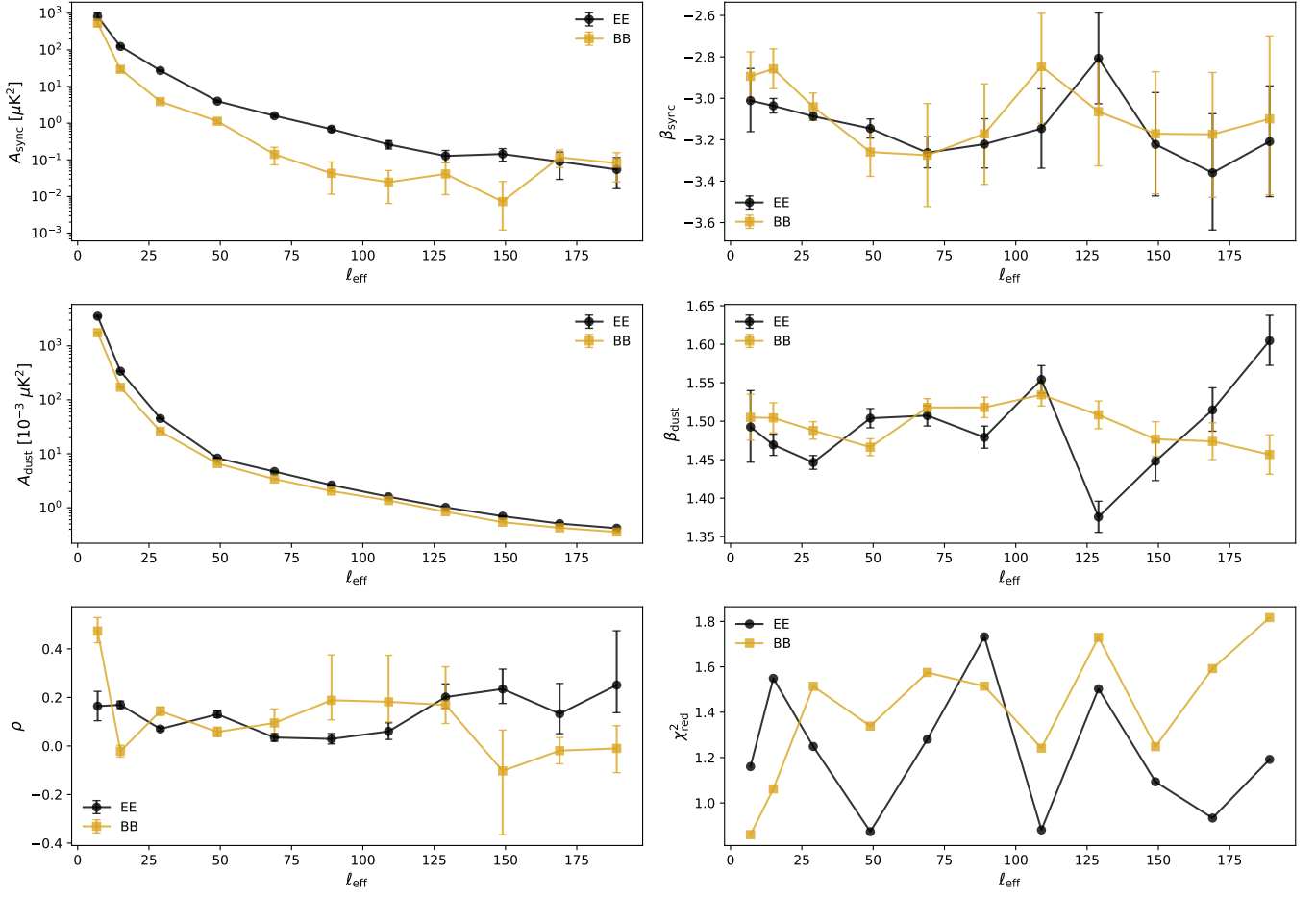


Figure 11: Bin-to-bin analysis results (observational) showing the evolution of fitted synchrotron and dust parameters as a function of multipole ℓ for both EE (top) and BB (bottom) modes.